What the probes read in the six Coulomb networks

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8 October 2026

1 What a minus means

Two different scores appear in the tables, and a minus means a different thing for each.

Probe score, written as R^2. The probe is a straight line fitted on the training systems and scored on the 4,000 validation systems.

$$R^2 = 1 - \frac{\text{squared error of the line}}{\text{squared error of guessing the validation average}}.$$

A score of 1 means the line reads the fact perfectly. A score of 0 means the line is no better than guessing the average value of that fact. A **negative** score means the line is worse than that guess. The training fit did not carry over, and its predictions on new systems are actively far from the labels. It does not mean the physical quantity itself is negative. The large entry -2.344 for potential energy in block 4 of the non-equivariant network is this kind of failure: the line misses by much more than the average would have.

Arrow score, written as cosine. Force direction is not an R^2. For each pair the model's own edge update is compared with the true Coulomb force:

$$\cos\theta = \frac{\text{update} \cdot \text{force}}{|\text{update}|\,|\text{force}|}.$$

The table reports the average of that cosine over pairs. $+1$ means the update points the same way as the force. 0 means the two arrows are unrelated. A **negative** cosine means the update points opposite the true force. -1 would be exactly backward. The shifted column in the raw logs is the same cosine after the forces are paired with the wrong system. A real alignment has to sit well above that shifted number.

2 What every network is given

None of the probed facts except the charges and the positions are handed to the network. Charge product, force magnitude, force direction, and potential energy are computed afterwards from the simulator, and the probe tries to read them back out.

The training target is the position of each particle after 1,000 simulator steps. The loss is the mean squared error between those positions and the network's guess. Lower is better. The graph layers calculate squared distance themselves by subtracting the two positions. They do not receive a distance matrix, a force matrix, or the potential energy.

Table 1: Inputs of one system of 10 particles. The edge table is not a force matrix. A 1 means the pair is allowed to interact. The diagonal is 0, so a particle does not interact with itself.

Array	Shape	Contents
Charge one-hot	10×2	$[1, 0]$ if charge -1, $[0, 1]$ if charge $+1$
Position	10×3	starting x, y, z
Velocity	10×3	starting velocity
Edges	10×10	1 for every pair, 0 on the diagonal

3 The six architectures

The three message-passing networks share one block. The drawings that follow mark the single change that makes each variant. In all three, and in the equivariant transformer, the positions themselves are not moved inside the block. The block adds a learned push into the velocity, and the guess is

$$\text{predicted position} = \text{initial position} + \text{updated velocity}.$$

That is one shot. It is not a replay of the 1,000 simulator steps. Widths below are the full models that produced the later tables: hidden width 64, message width 32, four blocks, except where a network is different.

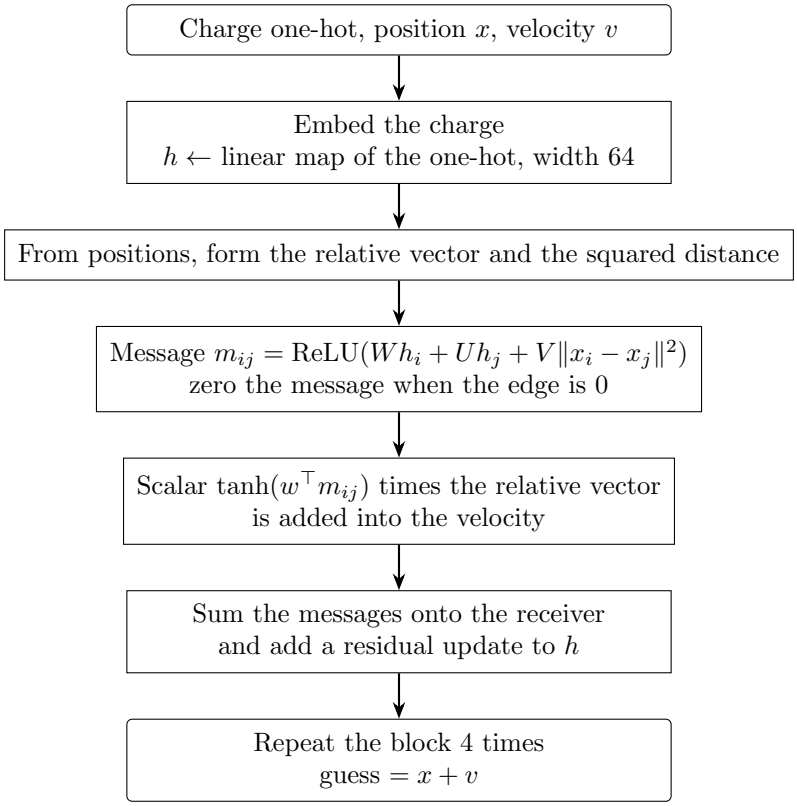


Figure 1: EGNN. The block is the Assignment 2 baseline. Distance enters only as a squared length, so the message is unchanged by a rotation of the whole system.

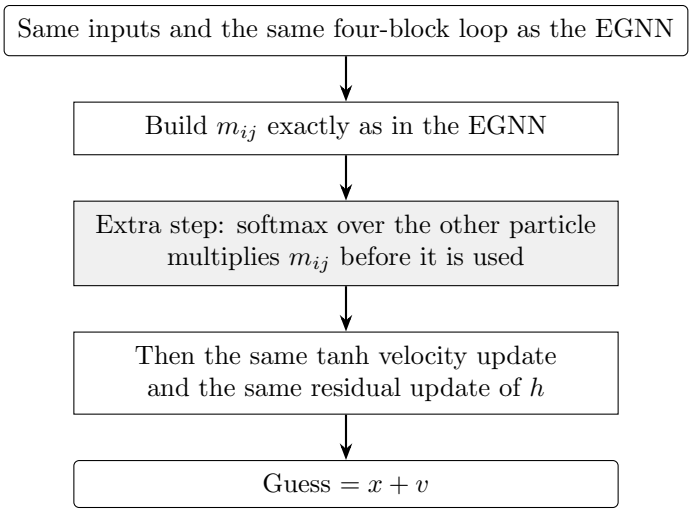


Figure 2: Attention EGNN. The shaded box is the only change. The attention weight depends on the message, which depends on charges and distance, so a rotation still does not change the message.

4 Position error through training

Each network was trained for 100 epochs on 2,400 systems and scored on 4,000 held-out systems. The number is the mean squared error of the final x, y, z. The logged epochs are 1, 20, 40, 60, 80, and 100.

Table 2: Validation position error. Attention EGNN is the most accurate. The Hamiltonian network's error rises after epoch 1. The MLP's training error at epoch 100 is 0.004, while the validation error stays near 0.22.

Network	1	20	40	60	80	100
Attention EGNN	0.669	0.029	0.021	0.018	0.016	0.014
Equivariant transformer	0.242	0.128	0.087	0.091	0.086	0.051
Non-equivariant GNN	0.741	0.101	0.083	0.081	0.080	0.079
EGNN	0.127	0.093	0.092	0.090	0.090	0.090
Hamiltonian network	0.083	0.101	0.113	0.121	0.117	0.126
MLP	0.138	0.210	0.231	0.229	0.224	0.222

5 Probe scores, block by block

The probe freezes the trained network, runs the validation systems through it, and fits a separate straight line from each layer onto one fact. Distance, charge product, and force magnitude are read from the pair message, one line for the 45 unordered pairs. Charge is read from the per-particle vector h. Potential energy is read from the average of the pair messages, one number per system. Force direction is the cosine defined above, not an R^2. Embed is the charge embedding before any block. Pair facts are not probed there, because the pair message does not exist yet. A blank cell means that measurement was not taken.

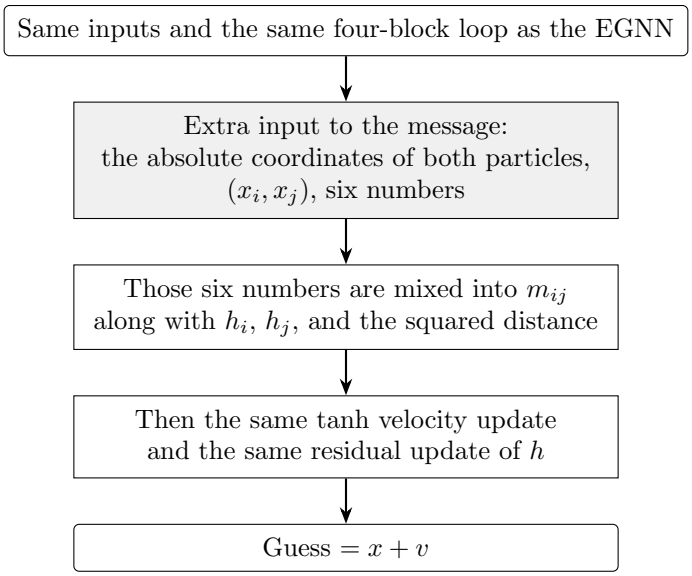


Figure 3: Non-equivariant GNN. The shaded box is the only change. Because the message can see where the particles sit in the coordinate frame, a rotation of the input no longer rotates the output in the same way.

The line is fit on the training systems only. The number in the table is the validation R^2, except the force rows, which are cosines.

Table 3: EGNN. Distance, charge product, force magnitude, and potential energy are strongest in block 1. Charge product and force magnitude fade in later blocks. The force arrow stays near cosine 0.

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.999	0.982	0.957	0.800
Charge product		0.997	0.362	0.432	0.132
Force magnitude		0.951	0.561	0.468	0.345
Potential energy		0.982	0.897	0.820	0.734
Force direction		0.017	−0.004	0.004	0.004

6 Deleting a high-scoring line

A line was removed only when its validation R^2 was at least 0.8. Force-direction rows were not removed, because they are not fitted lines. Removal means the component of the layer vector along that line is subtracted, and the same validation systems are run again. The position error is measured before and after. If the error rises, the guess was depending on that direction. If the error stays put, the fact was readable and this particular line was not needed for the guess.

Many removals with $R^2 \geq 0.8$ left the error unchanged to three digits. Table 9 keeps the removals that moved the validation error by at least 0.005.

The last two rows move the wrong way by a small amount: the error goes down. That is still

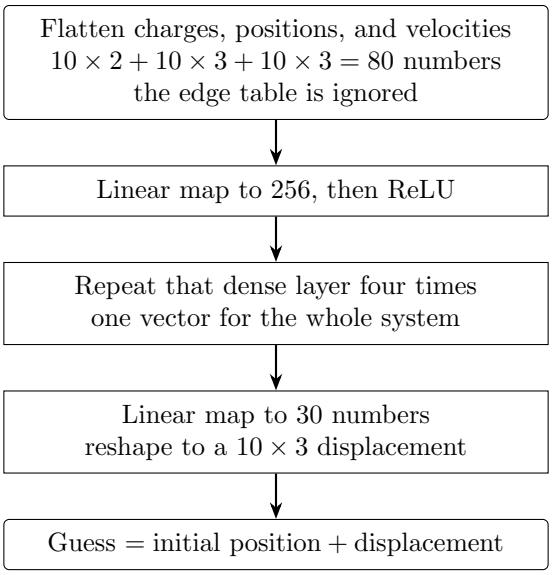


Figure 4: MLP. There is no pair message. The probe reads the single 256-dimensional vector and asks whether that one vector can recover every pair's fact at once.

Table 4: Attention EGNN. Charge is perfect in every layer. Distance and charge product peak in the middle of the network and stay below the EGNN. Potential energy is highest in block 2. The force arrow rises only to 0.084 by block 4.

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.397	0.741	0.759	0.502
Charge product		0.273	0.480	0.509	0.257
Force magnitude		0.813	0.645	0.706	0.788
Potential energy		0.588	0.761	0.700	0.742
Force direction		0.021	0.026	0.045	0.084

a change, but it is not evidence that the network needed the deleted line. The EGNN's perfect charge and distance scores are not in this table. Deleting those lines left the position error at 0.090.

7 Assignment 2

The assignment trained the same Coulomb task and reported validation error in the notebook discussion. Three of those runs correspond to networks here. The MLP, the Hamiltonian network, and the equivariant transformer were not in the assignment.

8 The same test on a much smaller copy

Each network was cut to one block and about eight times narrower, then trained and probed again. The EGNN in this copy has hidden width 8, message width 4, and 213 parameters. The full EGNN has four blocks of width 64. The position guesses get worse, except the small transformer, which

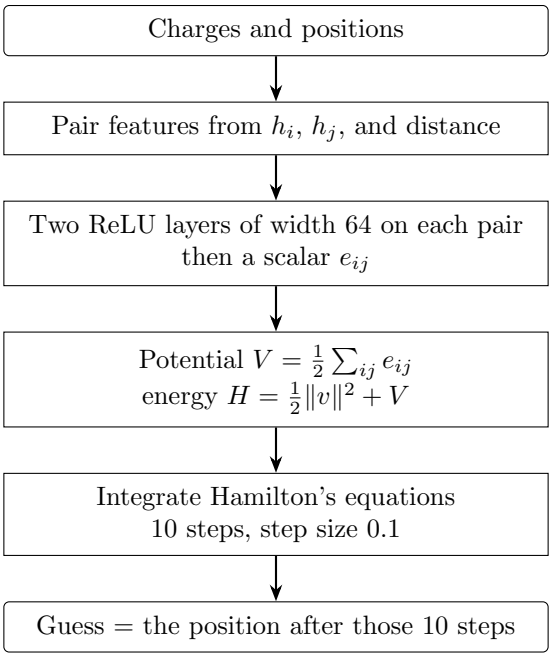


Figure 5: Hamiltonian network. The learned object is a scalar potential, not a direct position guess. Block 1 and block 2 in its probe table are the two ReLU layers of this potential, evaluated at the initial positions. The net-force cosine compares $-\partial H/\partial x$ at the start with the true net Coulomb force.

stays near its full-size error. Its force-arrow cosine is 0.79, the first time that arrow lines up, and deleting distance or the charge embedding makes the guess clearly worse.

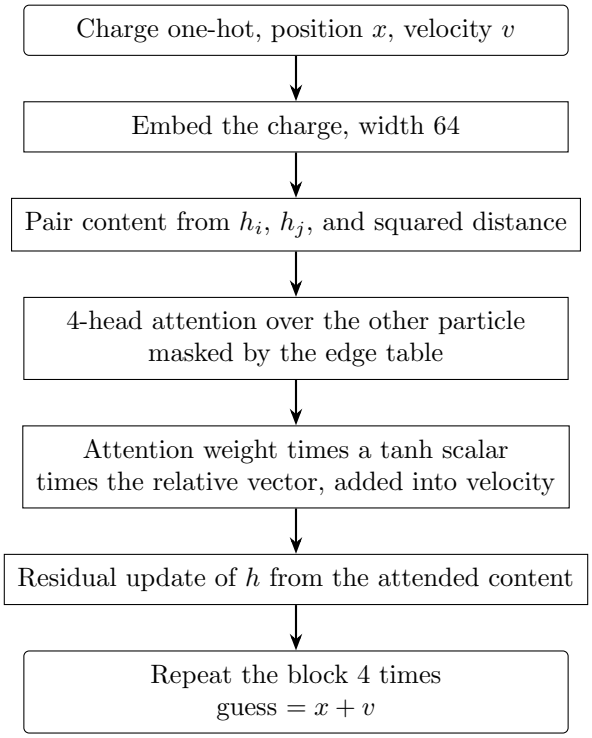


Figure 6: Equivariant transformer. This is not an Equiformer with irreducible representations. The features that attention mixes are invariant scalars. The arrow of the update still comes from the relative vector, scaled by those scalars.

Table 5: Non-equivariant GNN. Block 1 looks like the EGNN. Charge product has collapsed by block 4. Potential energy in block 4 is negative: the line is much worse than guessing the average. The force arrow stays near 0.

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	0.989	0.981
Distance		0.993	0.968	0.944	0.695
Charge product		0.915	0.233	0.091	0.003
Force magnitude		0.860	0.400	0.393	0.240
Potential energy		0.896	0.732	0.632	−2.344
Force direction		0.018	−0.004	0.004	−0.004

Table 6: Equivariant transformer. Distance stays near 0.94 in every block. Charge product does not appear. Force magnitude stays near 0.3, and potential energy only reaches 0.36 in block 4. The force arrow is strongest in block 1, at cosine 0.129, and fades.

Fact	Embed	Block 1	Block 2	Block 3	Block 4
Charge	1.000	1.000	1.000	1.000	1.000
Distance		0.937	0.938	0.941	0.943
Charge product		0.000	0.005	0.020	0.040
Force magnitude		0.264	0.267	0.274	0.304
Potential energy		0.000	0.034	0.092	0.360
Force direction		0.129	0.096	0.076	0.041

Table 7: Hamiltonian network. There is no block 3 or block 4. Block 1 and block 2 are the two layers of the learned potential, at the initial positions. Charge product, force magnitude, and potential energy appear in block 2. The net-force row is a cosine, not an R^2.

Fact	Embed	Block 1	Block 2
Charge	1.000		
Distance		1.000	1.000
Charge product		0.012	1.000
Force magnitude		0.650	0.938
Potential energy		0.056	0.887
Net force			0.085

Table 8: MLP. There is no pair message and no charge embedding before the first dense layer. Charge is readable in block 1 and fades. Distance, charge product, force magnitude, and potential energy stay near 0 or below. A negative entry means the line is worse than guessing the average.

Fact	Block 1	Block 2	Block 3	Block 4
Charge	0.909	0.733	0.520	0.345
Distance	0.191	0.106	0.038	−0.013
Charge product	−0.007	−0.033	−0.051	−0.062
Force magnitude	−0.024	−0.048	−0.061	−0.075
Potential energy	−0.026	−0.013	−0.065	−0.178

Table 9: Deletions that changed the position error. The largest change is force magnitude in block 1 of the non-equivariant network.

Network	Line removed	Before	After
Non-equivariant GNN	Force magnitude, block 1	0.079	0.458
Hamiltonian network	Distance, block 2	0.126	0.323
Equivariant transformer	Distance, block 1	0.051	0.216
Non-equivariant GNN	Distance, block 1	0.079	0.114
Non-equivariant GNN	Charge product, block 1	0.079	0.102
Attention EGNN	Charge, block 2	0.014	0.028
EGNN	Charge product, block 1	0.090	0.103
Hamiltonian network	Distance, block 1	0.126	0.137
Attention EGNN	Charge, embedding	0.014	0.019
MLP	Charge, block 1	0.222	0.212
Hamiltonian network	Charge, embedding	0.126	0.120

Table 10: Assignment 2 validation error next to the epoch-100 error of the closest network in this project.

Assignment 2 run	Network here	Assignment 2	Here
Attention	Attention EGNN	0.0103	0.014
Full EGNN	EGNN	0.0945	0.090
No equivariance	Non-equivariant GNN	0.0579	0.079
No residual	—	0.0896	—
No velocity	—	0.0983	—
No distance	—	0.1279	—
No tanh	—	11.46	—

Table 11: Small networks, one block. Position error is the validation mean squared error at epoch 100. The other columns are probe R^2, except the force arrow, which is a cosine. The Hamiltonian net's arrow column is the net-force cosine.

Network	Error	Charge	Distance	$q_i q_j$	$\|F\|$	Potential	Arrow
EGNN	0.129	1.00	0.22	0.00	0.03	0.00	−0.23
Attention EGNN	0.125	1.00	−1.23	0.28	0.25	−4.65	0.39
Non-equivariant GNN	0.131	1.00	0.01	0.00	0.01	−0.01	0.00
Transformer	0.085	1.00	0.94	0.00	0.26	0.00	0.79
Hamiltonian net	0.125	1.00	0.91	0.99	0.48	0.75	0.21
MLP	0.140	0.04	0.00	0.00	−0.01	−0.01	—

Table 12: Deletions on the small transformer, the only small network whose position error stays near the full model. Both deletions make the guess worse.

Line removed	Before	After
Charge, embedding	0.085	0.298
Distance, block 1	0.085	0.245